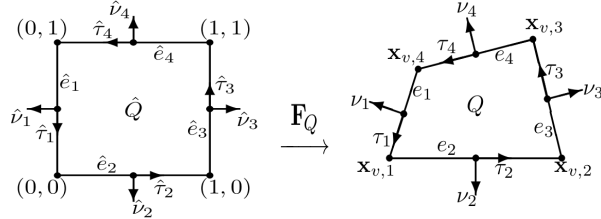


A Modified Bernardi-Raugel Element using Direct Serendipity Space

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Quadrilateral Mesh



Question

Can we use Bernardi-Raugel element on quadrilateral mesh directly without mapping?

First order Direct Serendipity Space

Auxiliary functions

We index the edges e_i in the counterclockwise direction, and define the outer unit normal as ν_i , tangents as τ_i and vertices as $\mathbf{x}_{v,i}$, then define special linear function λ_i for $i = 1, 2, 3, 4$.

$$\lambda_i(\mathbf{x}) = -(\mathbf{x} - \mathbf{x}_{v,i}) \cdot \nu_i, \quad i = 1, 2, 3, 4,$$

Then define

$$R_{13}(\mathbf{x}) = \frac{\lambda_1 - \lambda_3}{\lambda_1 + \lambda_3}, \quad R_{24}(\mathbf{x}) = \frac{\lambda_2 - \lambda_4}{\lambda_2 + \lambda_4},$$

and

$$R_1(\mathbf{x}) = \frac{1}{2}(1 - R_{13}), \quad R_3(\mathbf{x}) = \frac{1}{2}(1 + R_{13}), \\ R_2(\mathbf{x}) = \frac{1}{2}(1 - R_{24}), \quad R_4(\mathbf{x}) = \frac{1}{2}(1 + R_{24}).$$

Function R_i is 1 on corresponding edge e_i , 0 on the opposite edge, and arbitrary on two neighbours.

$$\mathbb{S}_1(Q) = \text{span}\{\lambda_2\lambda_4R_{13}, \lambda_1\lambda_3R_{24}\},$$

Resulted $\mathbb{DS}_1(Q)$ space

$$\mathbb{DS}_1(Q) = \mathbb{P}_1 \oplus \mathbb{S}_1(Q),$$

Defined on the given quadrilateral Q .

Modified Bernardi-Raugel finite element

The modified Bernardi-Raugel finite element is therefore defined as following:

$$\mathbb{V}_{S,h}(Q) = \mathbb{DS}_1^2(Q) \oplus \text{span}\{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3, \mathbf{b}_4\}$$

$$\mathbb{W}_{D,h}(Q) = \mathbb{W}_1^0(Q) = \mathbb{P}_0,$$

Edge basis

Four supplemental functions are defined modulo 4 by

$$\mathbf{b}_i(\mathbf{x}) = \frac{\varphi_i(\mathbf{x})}{\varphi_i(\mathbf{x}_i)} \nu_i, \quad \mathbf{x}_i = \frac{1}{2}(\mathbf{x}_{v,i-1} + \mathbf{x}_{v,i}), \quad \varphi_i = \lambda_{i+1}\lambda_{i+3}R_i,$$

We remark that the scalar function ϕ_i lie in second order serendipity space $\mathbb{DS}_2(Q)$.

Nodal basis

Set auxiliary rational function first

$$r(\mathbf{x}) = \frac{\lambda_3(\mathbf{x})\lambda_4(\mathbf{x})}{\lambda_3(\mathbf{x}_{v,1})\lambda_4(\mathbf{x}_{v,1})} - \frac{\lambda_1(\mathbf{x})\lambda_4(\mathbf{x})}{\lambda_1(\mathbf{x}_{v,2})\lambda_4(\mathbf{x}_{v,2})} + \\ \frac{\lambda_1(\mathbf{x})\lambda_2(\mathbf{x})}{\lambda_1(\mathbf{x}_{v,3})\lambda_2(\mathbf{x}_{v,3})} - \frac{\lambda_2(\mathbf{x})\lambda_3(\mathbf{x})}{\lambda_2(\mathbf{x}_{v,4})\lambda_3(\mathbf{x}_{v,4})}$$

and then define

$$R(\mathbf{x}) = r(\mathbf{x}) - \sum_{i=1}^4 r(\mathbf{x}_{e,i}) \frac{\varphi_i(\mathbf{x})}{\varphi_i(\mathbf{x}_i)}.$$

Let $\nu_{d,i}$ be a unit normal to the first diagonal joining $\mathbf{x}_{v,i}$ with $\mathbf{x}_{v,i+2}$, and define new linear functions as

$$\lambda_{d,1} = -(\mathbf{x} - \mathbf{x}_{v,1}) \cdot \nu_{d,1} \quad \text{and} \quad \lambda_{d,2} = -(\mathbf{x} - \mathbf{x}_{v,2}) \cdot \nu_{d,2}.$$

Then the four nodal basis functions are defined as following

$$\psi_{v,1}(\mathbf{x}) = \frac{\lambda_{d,2}(\mathbf{x}) - \frac{1}{2}\lambda_{d,2}(\mathbf{x}_{v,3})(1 + R(\mathbf{x}))}{\lambda_{d,2}(\mathbf{x}_{v,1}) - \lambda_{d,2}(\mathbf{x}_{v,3})}, \\ \psi_{v,2}(\mathbf{x}) = \frac{\lambda_{d,1}(\mathbf{x}) - \frac{1}{2}\lambda_{d,1}(\mathbf{x}_{v,4})(1 + R(\mathbf{x}))}{\lambda_{d,1}(\mathbf{x}_{v,2}) - \lambda_{d,1}(\mathbf{x}_{v,4})}, \\ \psi_{v,3}(\mathbf{x}) = \frac{\lambda_{d,2}(\mathbf{x}) - \frac{1}{2}\lambda_{d,2}(\mathbf{x}_{v,1})(1 + R(\mathbf{x}))}{\lambda_{d,2}(\mathbf{x}_{v,3}) - \lambda_{d,2}(\mathbf{x}_{v,1})}, \\ \psi_{v,4}(\mathbf{x}) = \frac{\lambda_{d,1}(\mathbf{x}) - \frac{1}{2}\lambda_{d,1}(\mathbf{x}_{v,2})(1 + R(\mathbf{x}))}{\lambda_{d,1}(\mathbf{x}_{v,4}) - \lambda_{d,1}(\mathbf{x}_{v,2})}.$$

An alternative definition of $R(\mathbf{x})$

Redefine as modulo 4

$$R_{i,i+1} = \frac{\lambda_{i-2}\lambda_{i-1}}{(\lambda_{i-2} + \alpha\lambda_i)(\lambda_{i-1} + \beta\lambda_{i+1})}$$

where

$$\alpha = -\frac{\tau \cdot \nu_i}{\tau \cdot \nu_{i+1}}, \quad \beta = -\frac{(\mathbf{x}_{v,i} - \mathbf{x}_{v,i+1}) \cdot \nu_{i-2}}{(\mathbf{x}_{v,i} - \mathbf{x}_{v,i+1}) \cdot \nu_{i-1}}$$

Normalized as

$$\tilde{R}_{i,i+1} = \frac{R_{i,i+1}}{\sum_{k=1}^4 R_{k,k+1}}$$

Then the alternative $R(\mathbf{x})$ is defined

$$R = \tilde{R}_{1,2} + \tilde{R}_{3,4} - \tilde{R}_{1,3} - \tilde{R}_{2,4}$$

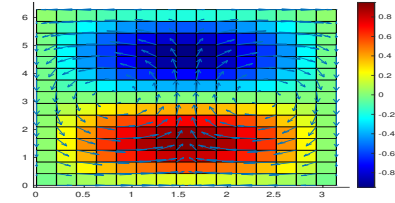
Preliminary numerical results

Stokes equation with exact following solution

$$u = \cos(x) \sin(y), \quad v = -\sin(x) \cos(y)$$

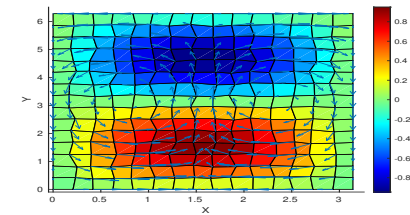
$$\frac{\partial p}{\partial x} = \cos(x) \sin(y), \quad \frac{\partial p}{\partial y} = \sin(x) \cos(y)$$

Rectangular Mesh



Grid size	16 × 16	32 × 32	64 × 64
$L^2(\text{res}_u)$	1.052×10^{-1}	2.513×10^{-2}	6.130×10^{-3}
$L^2(\text{res}_p)$	4.463×10^{-1}	2.093×10^{-1}	1.019×10^{-1}
$L^2(\text{res}_{du})$	5.845×10^{-1}	1.374×10^{-1}	3.330×10^{-2}

Quadrilateral Mesh



Grid size	16 × 16	32 × 32	64 × 64
$L^2(\text{res}_u)$	1.074×10^{-2}	2.567×10^{-2}	6.266×10^{-3}
$L^2(\text{res}_p)$	4.670×10^{-1}	2.207×10^{-1}	1.078×10^{-1}
$L^2(\text{res}_{du})$	5.981×10^{-1}	1.525×10^{-1}	4.580×10^{-2}

Summary

1. Achieves second order convergence on velocity, first order convergence on derivatives.
2. Same convergence rate achieved on both rectangular and quadrilateral mesh.